

Deel

Marketing Analytics Challenge

Executive Presentation | Tasks 1, 2 & 3

Period: Jul 2023 – Jun 2025 | 11,952 records | 7 teams | 17 programs

Tools: Python · Pandas · NumPy · Plotly · Streamlit · Matplotlib · python-pptx

Code Python

[Link to Github](#)

Dashboard Executive

[Link to Extra Insights](#)

Task 1

Sales Funnel Analysis

Task 2

Paid & Brand Channels

Task 3

Scenario Analysis

Who You're Presenting To

Two senior stakeholders with distinct focus areas

Paid Marketing Director

Cares about short-term ROI and performance of paid spend.

Key Questions:

- Which channels generate the best CPA?
- Where is budget being wasted?
- How do we grow Won at lower cost?

Head of Marketing

Wants to prove long-term impact of awareness and brand-building.

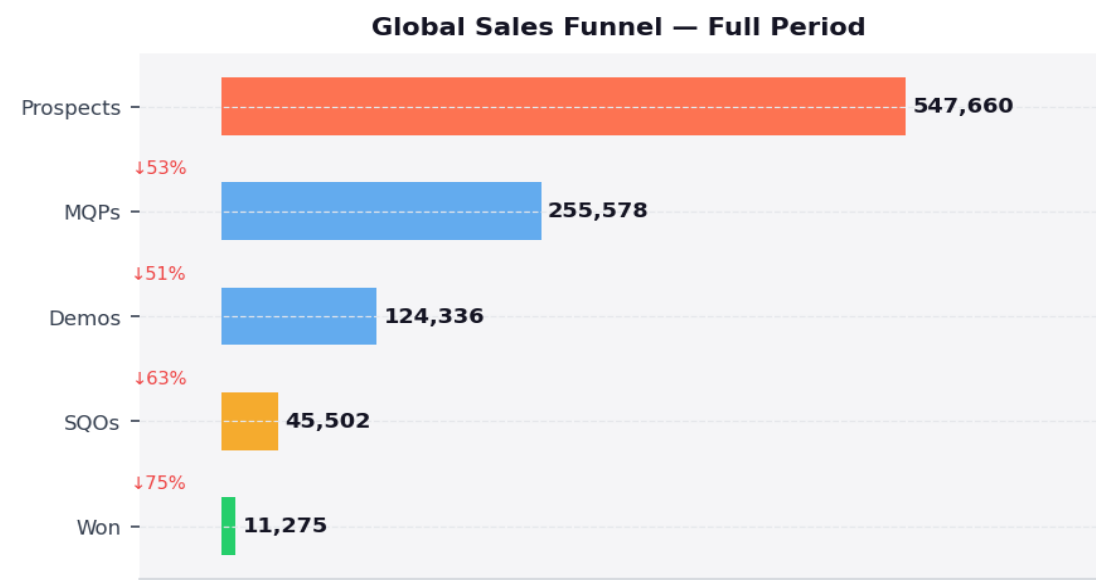
Key Questions:

- Are organic channels growing over time?
- What is the implied value of zero-cost channels?
- How do we measure brand budget impact?

Portfolio Overview — Key Metrics at a Glance

Full period Jul 2023 – Jun 2025 | All channels combined

Total Investment	Total Won Opps.	Avg. Global CPA	E2E Conv. Rate
\$422K	11,275	\$37	2.06%
Total Impressions	Total Clicks	Total Demos	Total SQOs
11.2B	70.9M	124,336	45,502



Funnel Drop-off Summary		
Prospects → MQPs	46.7%	filter applied at top of funnel
MQPs → Demos	48.6%	quality gate
Demos → SQOs	36.6%	sales qualification
SQOs → Won	24.8%	deal closure
Overall E2E	2.06%	end-to-end conversion

TASK 1

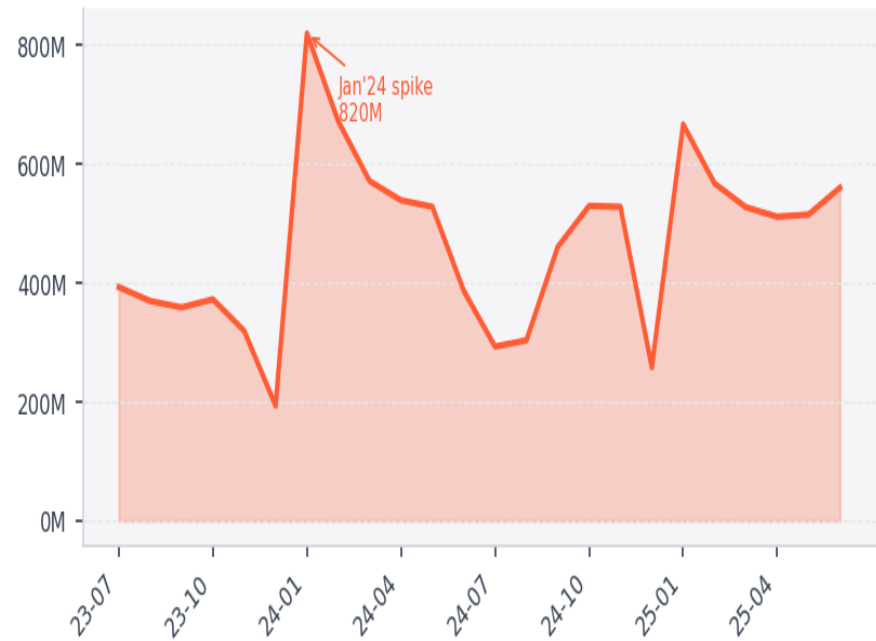
Analyzing Sales Funnel Data

Patterns over time · Seasonality · Channel conversion rates · Recommendations

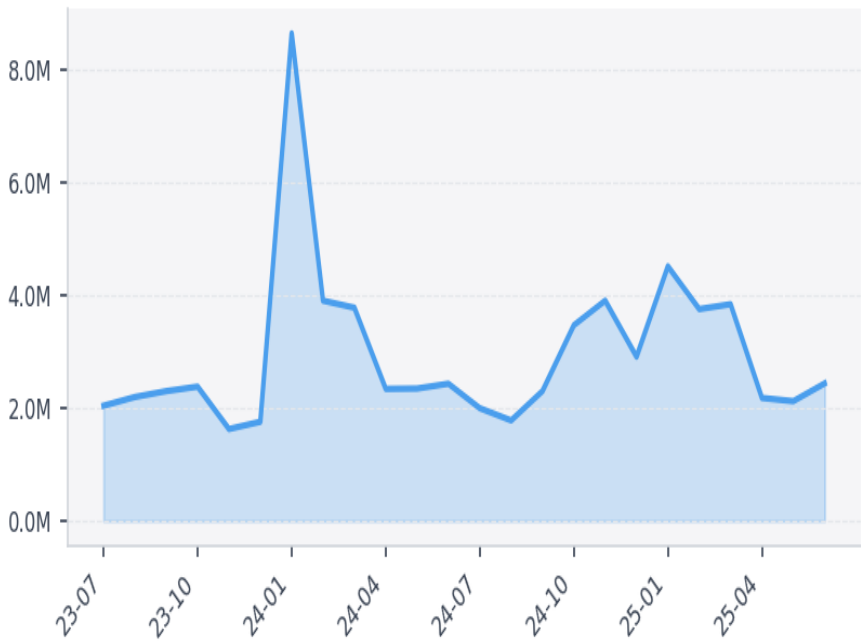
Task 1 — Top-Funnel: Impressions & Clicks Over Time

Awareness stage metrics | Jul 2023 – Jun 2025

Impressions Over Time



Clicks Over Time



Total Impressions

11.2B

Total Clicks

70.9M

Overall CTR

0.63%

Observations

Jan 2024 spike: 820M impressions vs ~380M avg — likely a large awareness campaign launch.

Click volume grew steadily from Jul 2023 to Oct 2024, then plateaued — market saturation signal.

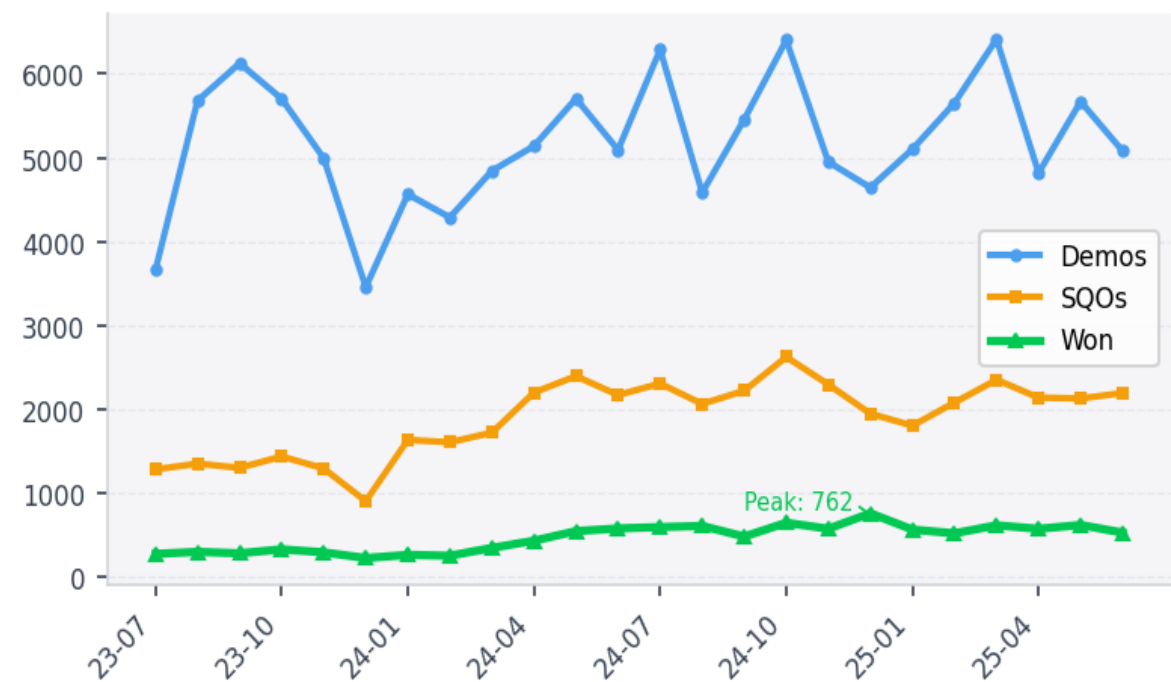
CTR of 3.64% for paid_search is strong; paid_social at 0.47% signals poor audience fit.

Impressions and clicks are concentrated in the paid team (~6K of 11.9K rows); organic channels have no impression data.

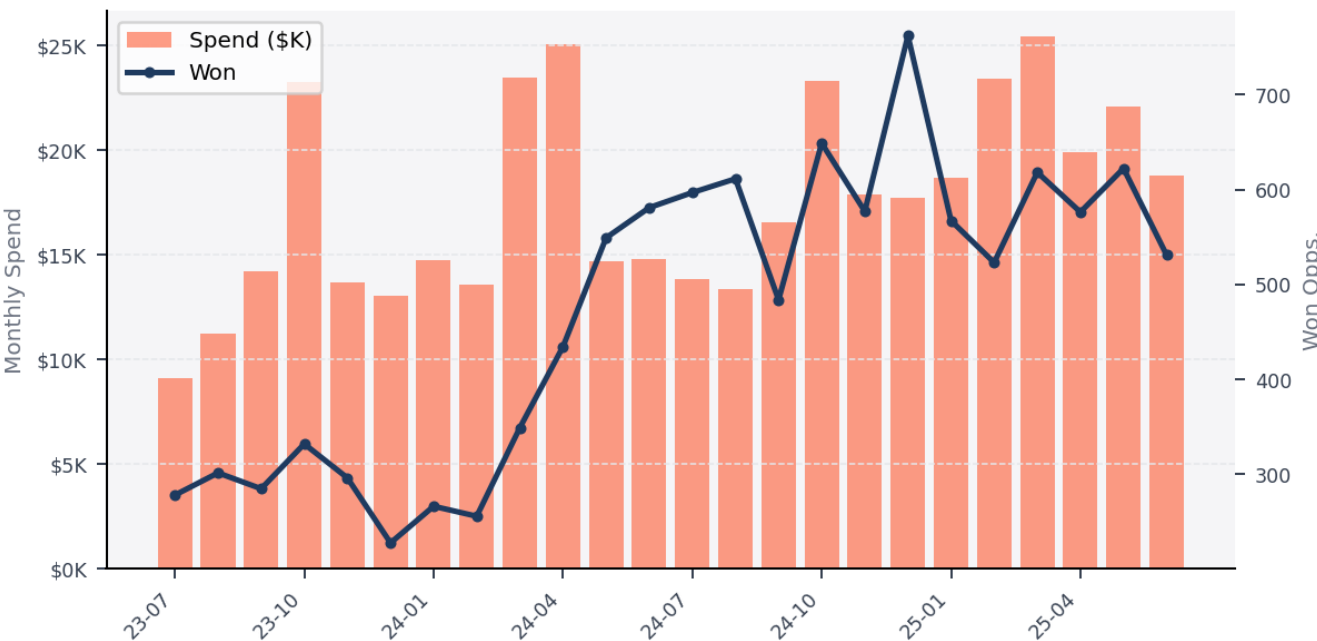
Task 1 — Bottom-Funnel: Demos, SQOs & Won Over Time

Conversion outcomes | Jul 2023 – Jun 2025

Bottom-Funnel Volumes Over Time



Monthly Spend vs Won



Q4 Surge (Oct–Dec)

Enterprise year-end budget decisions accelerate; Y2 peaked at 622 Won in May. Nov–Dec show consistent uplift across both cycles.

Jan–Feb Dip

Budget resets slow deal closures. Jan 2025 dropped to ~350 Won despite high spend — CPA spikes to \$50–65K in this period.

Upward Trend

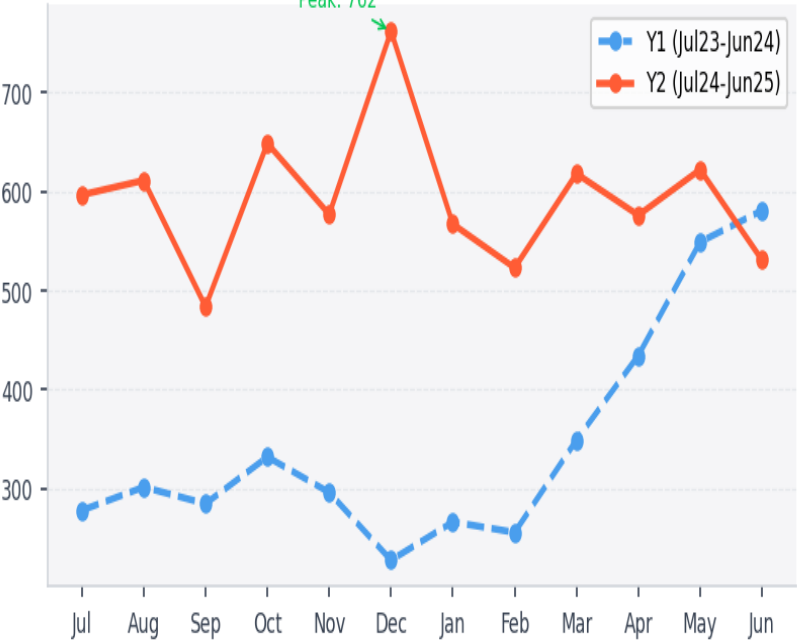
Total Won grew from ~285/mo (Y1 avg) to ~580/mo (Y2 avg) — a 2x improvement driven by paid scale-up and lifecycle growth.

Task 1 — Seasonality Patterns by Fiscal Cycle

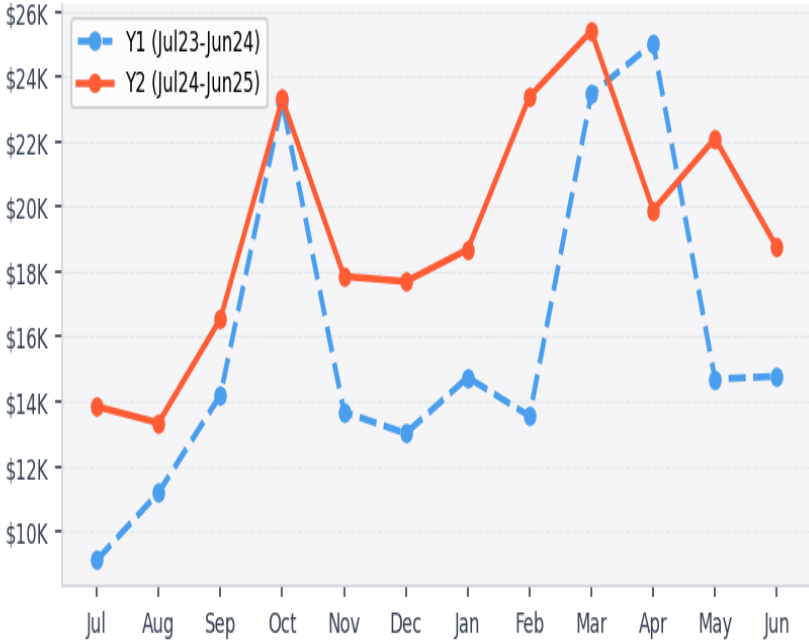
Fiscal year: Jul (Month 1) → Jun (Month 12)

Won by Fiscal Month — YoY

Peak: 762



Investment by Fiscal Month — YoY



Seasonality Drivers

Oct–Dec Peak

Enterprise fiscal year-end. Companies accelerate buying decisions before Dec close.

May–Jun Peak (Y2)

Strong Q4 performance repeated — suggests maturing sales cycle management.

Jan–Mar Slow

Budget resets. Decision-makers pause. CPA rises 30–50% vs Q4 levels.

Jan'24 Spend Spike

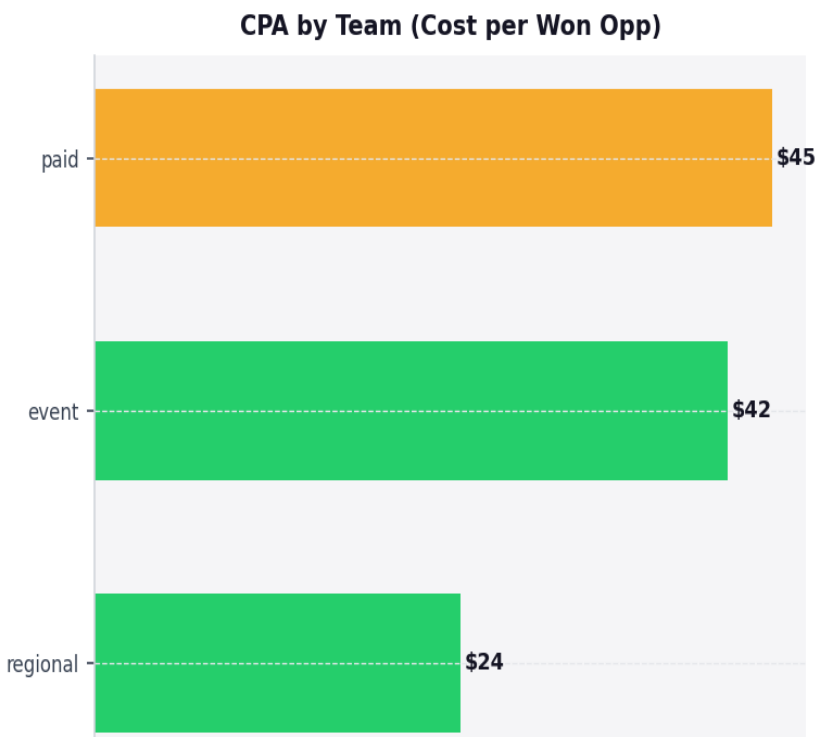
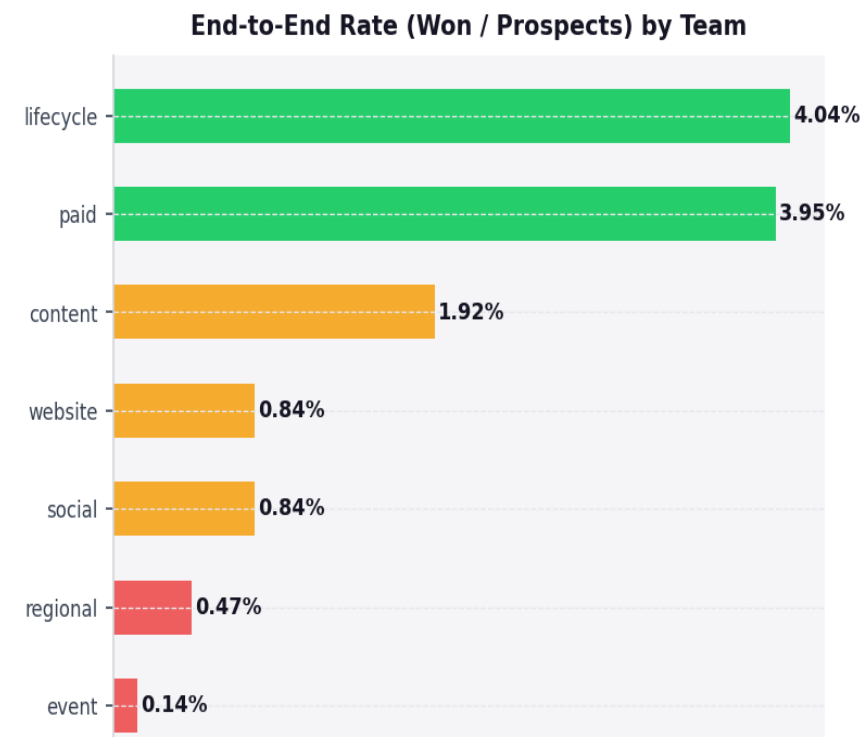
Investment jumped 3x in Jan 2024 — aggressive push against Jan dip; limited Won impact.

Y2 Consistently Higher

Won per month up 2x vs Y1 across all months, driven by lifecycle and paid optimization.

Task 1 — Funnel Efficiency by Marketing Team

Which teams convert best? $E2E\ Rate = Won / Prospects$



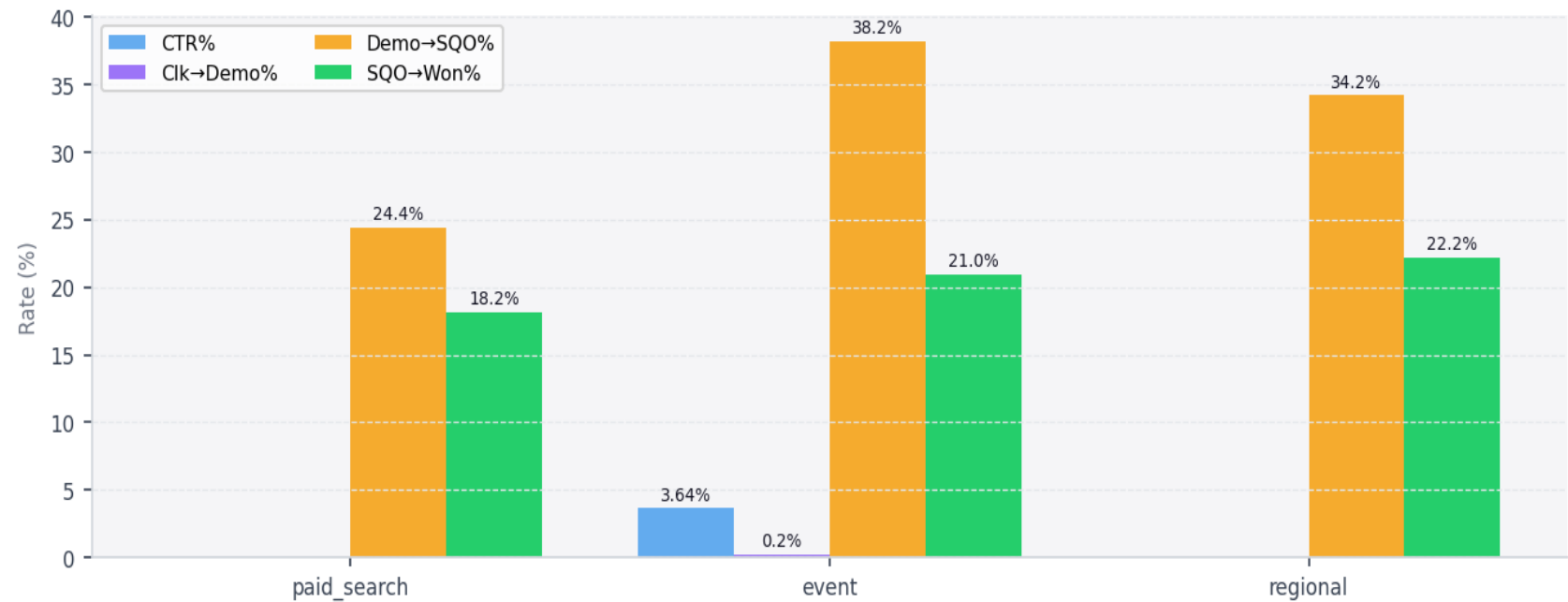
Lifecycle leads with 4.0% E2E rate (highest of all teams) at ZERO cost. Paid has the most volume but only 0.13% E2E — 31x less efficient than lifecycle per prospect. Regional outperforms paid at the lowest CPA in the portfolio with far less investment.

Team	Won	E2E%	CPA
lifecycle	686	4.04%	—
paid	8,942	3.95%	\$45
content	874	1.92%	—
website	248	0.84%	—
social	39	0.84%	—
regional	250	0.47%	\$24
event	236	0.14%	\$42

Task 1 — 3-Channel Conversion Rates

Paid Search · Event · Regional | Impression → Click → Demo → SQO → Won

Conversion Rates: Paid Search vs Event vs Regional



Regional is the standout: lowest CPA (\$17K vs \$47K paid search) with a strong 34% Demo→SQO rate, yet received only 1.2% of paid budget. Event drives quality demos (24% Demo→SQO) with zero impression cost. Paid search volume leader but efficiency gap vs regional is significant.

Metric	Paid Search	Event	Regional
CTR%	3.64%	n/a	n/a
Clk→Demo%	0.21%	n/a	n/a
Demo→SQO%	38.2%	24.4%	34.2%
SQO→Won%	21.0%	18.2%	22.2%
CPA (\$)	\$45.52	\$42.33	\$24.46
Total Spend	\$132K	\$10K	\$6K

Task 1 — Recommendations: Improve Funnel Efficiency & Volume

1

Reallocate paid_social → paid_brand_search

Paid social CPA is \$197K — 11× worse than brand search (\$18K). Redirecting 30% of paid social budget (~\$24M) to brand search could generate +1,300 additional Won per year at the same total spend.

2

Scale regional demand gen (under-funded)

Regional has the best CPA (\$17K) and strong Demo→SQO (34%) yet only received \$4.3M total (1.2% of paid budget). 3× budget increase to \$13M could yield ~750 additional Won.

3

Invest in lifecycle / CRM automation

Lifecycle holds the highest E2E rate (4.04%) at zero media cost. Investing in Braze sequence expansion and lead scoring would capture warm prospects that currently leak or convert via expensive paid.

4

Protect Q4 spend; reduce Q1 scale

Won peaks in Oct–Dec and May–Jun with strong CPA. Jan–Mar shows worst CPA (\$50–65K). Shift Q1 budget to lifecycle and content to maintain pipeline at lower cost during slow conversion period.

Task 1 — Seasonality: Patterns, Root Causes & Implications

Why do metrics rise and fall? Business reasons behind the data

Pattern Observed	Probable Cause	Implication for Marketing
Impressions spike in Jan 2024 (3× above period average)	Deliberate large-scale awareness campaign launched by the paid team — not organic seasonality	Isolate campaign periods before attributing results to market behaviour
Won drops every Jan–Mar (consistent both years)	B2B buyers reset annual budgets in January; decision-making slows across the industry	Avoid scaling paid CPA spend in Q1 — it's the most expensive period per Won
Won peaks Oct–Dec (Q4 enterprise surge)	Enterprise customers close deals before their own fiscal year-end to spend allocated budgets	Front-load pipeline building in Aug–Sep to harvest demand when buyers are ready
Won peaks again in May–Jun (second seasonal high)	Deals entered in Nov–Dec mature after ~150-day avg sales cycle, closing in May–Jun	Feed top-of-funnel aggressively in Nov so pipeline ripens for the May peak
CTR declining across 2024–2025 (paid_search & paid_social)	Audience saturation and creative fatigue — same users seeing repetitive ads, ignoring them	Rotate creatives, expand audience targeting, and shift budget to brand search where CTR is 6.7%

TASK 2

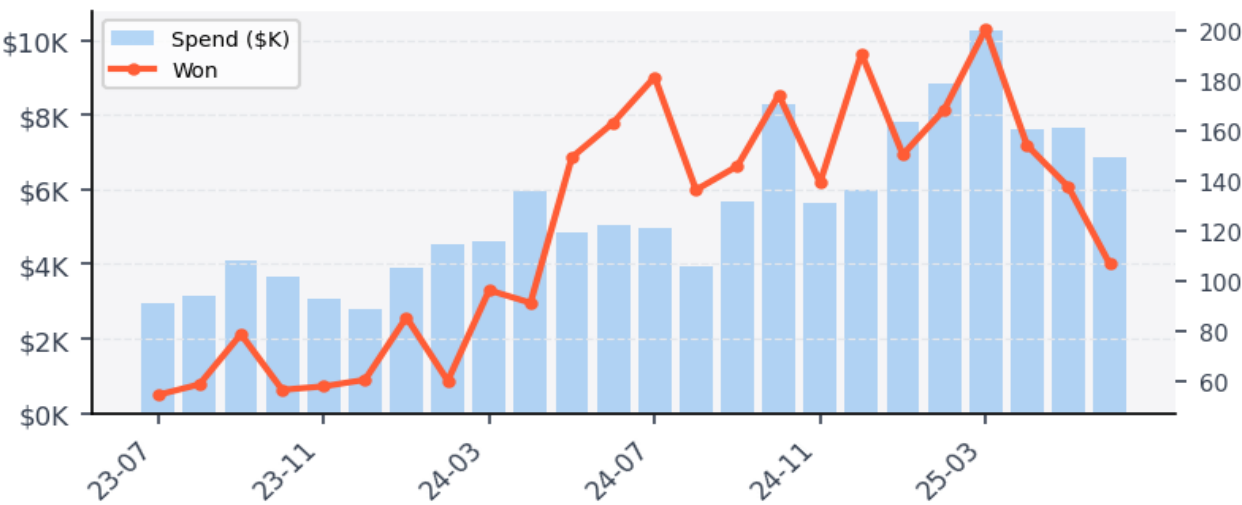
Paid & Brand Marketing Channel Analysis

Cost vs outcomes · CPM/CTR efficiency · Brand growth · Budget reallocation

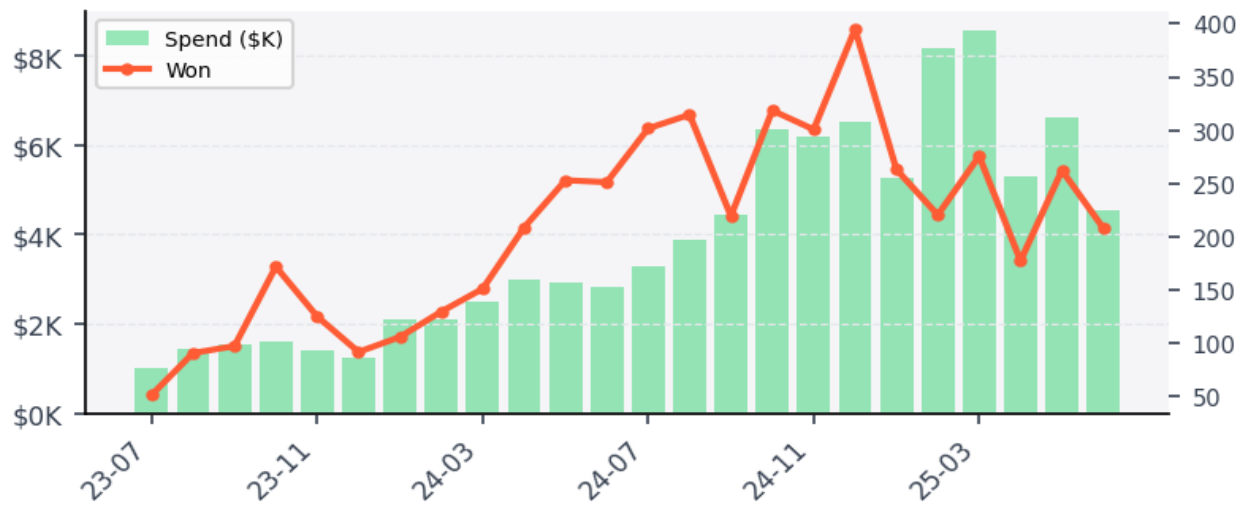
Task 2 — Paid Search vs Brand Search: Cost → Demos → SQOs → Won

Two core paid channels analyzed across the full period

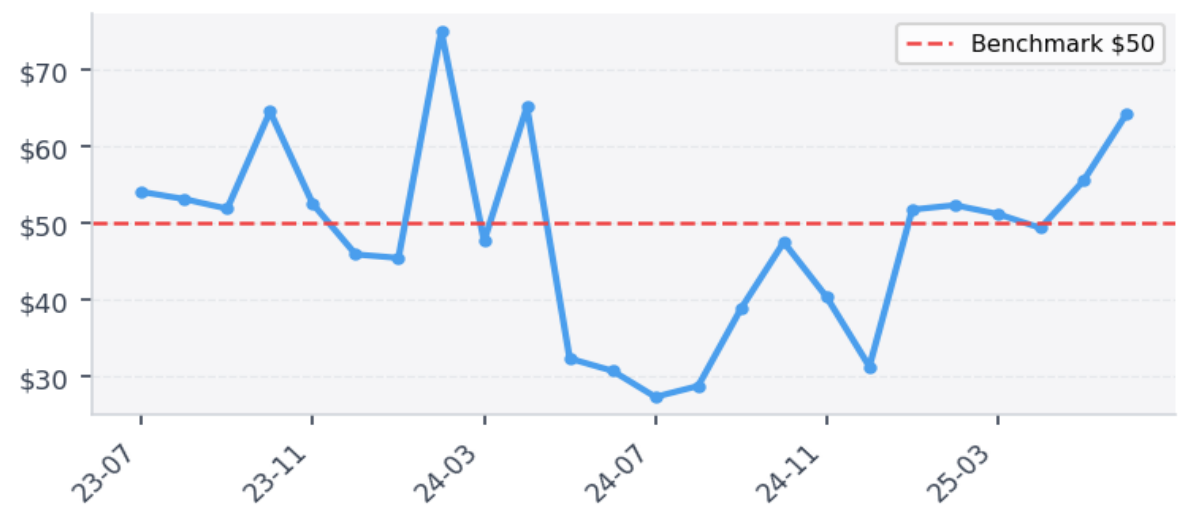
Paid Search — Spend vs Won



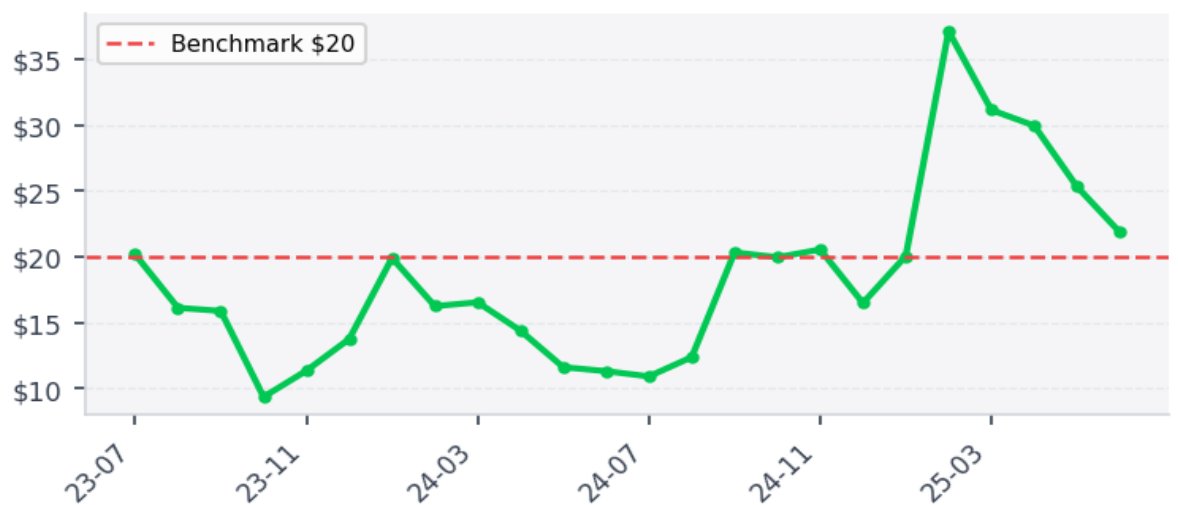
Brand Search — Spend vs Won



Paid Search — CPA Trend

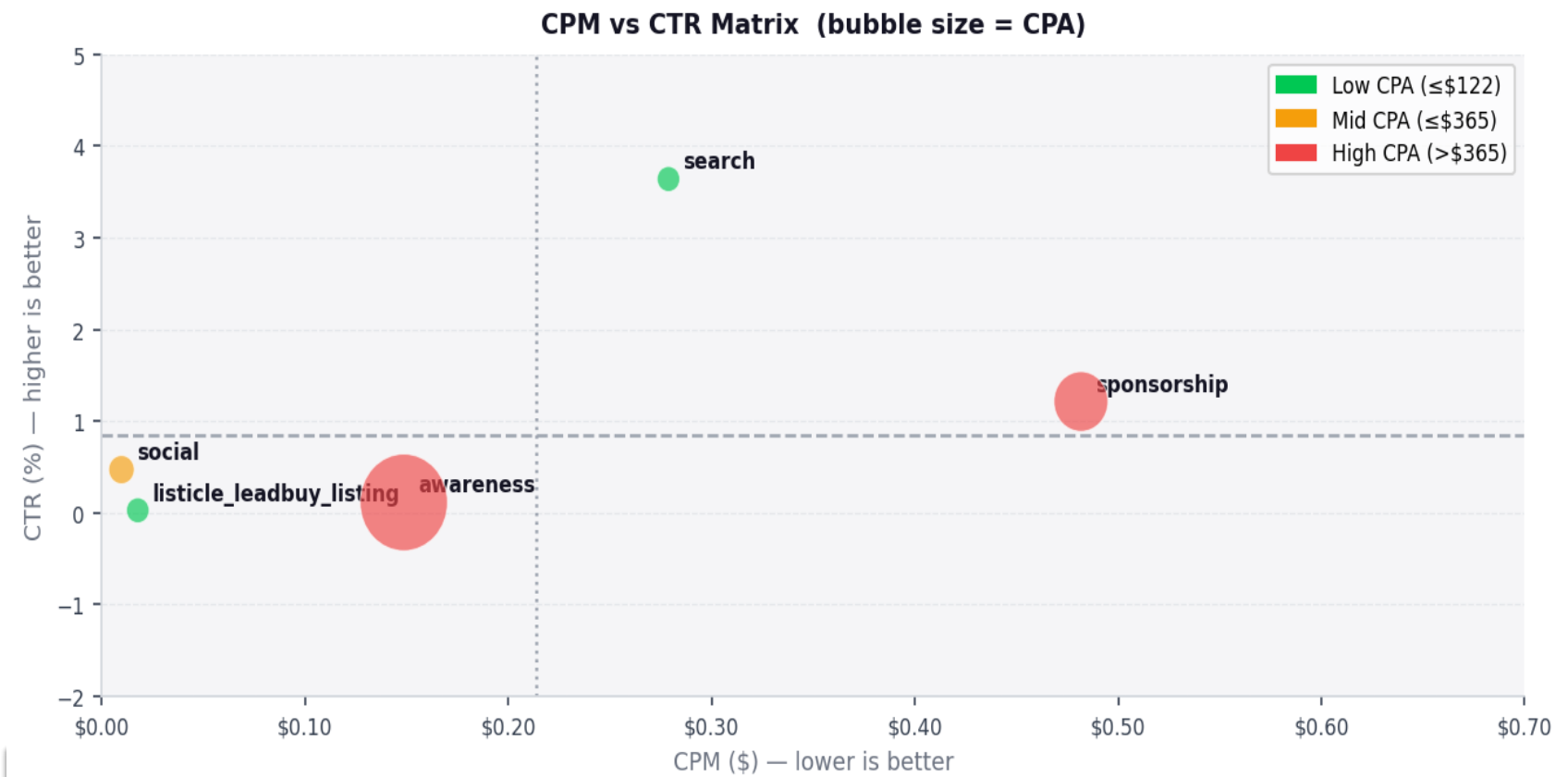


Brand Search — CPA Trend



Task 2 — CPM vs CTR Matrix: Paid Channel Efficiency

Upper-left quadrant (low CPM, high CTR) = most efficient awareness spend



conversions). Paid social shows 0.47% CTR despite being the highest-spend program — clear candidate for budget cut.

Channel Efficiency Summary

Highest CTR

paid_brand_search 6.68%

Lowest CPM

paid_social \$0.0096

Best CPA

paid_brand_search \$18.67

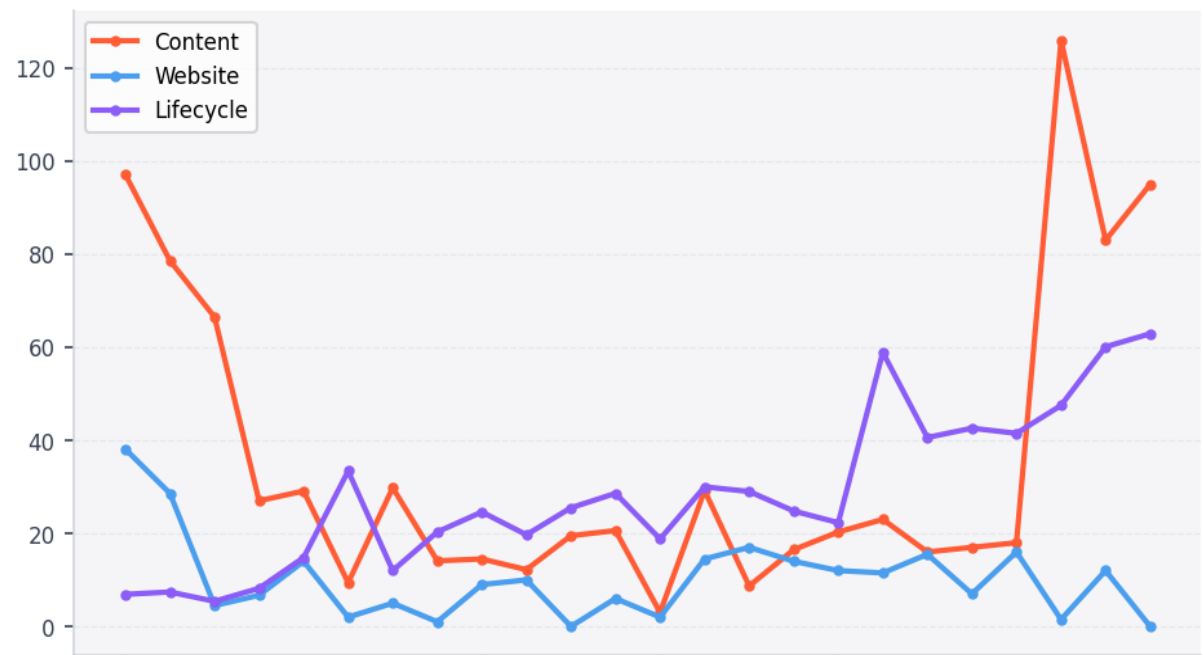
Worst CPA

paid_awareness \$7,932

Task 2 — Brand Channels: Content, Website & Lifecycle Growth

Zero-cost channels building long-term equity

Zero-Cost Channels — Monthly Won



Content Won
Y1→Y2

418→456

+9%

Website Won
Y1→Y2

125→123

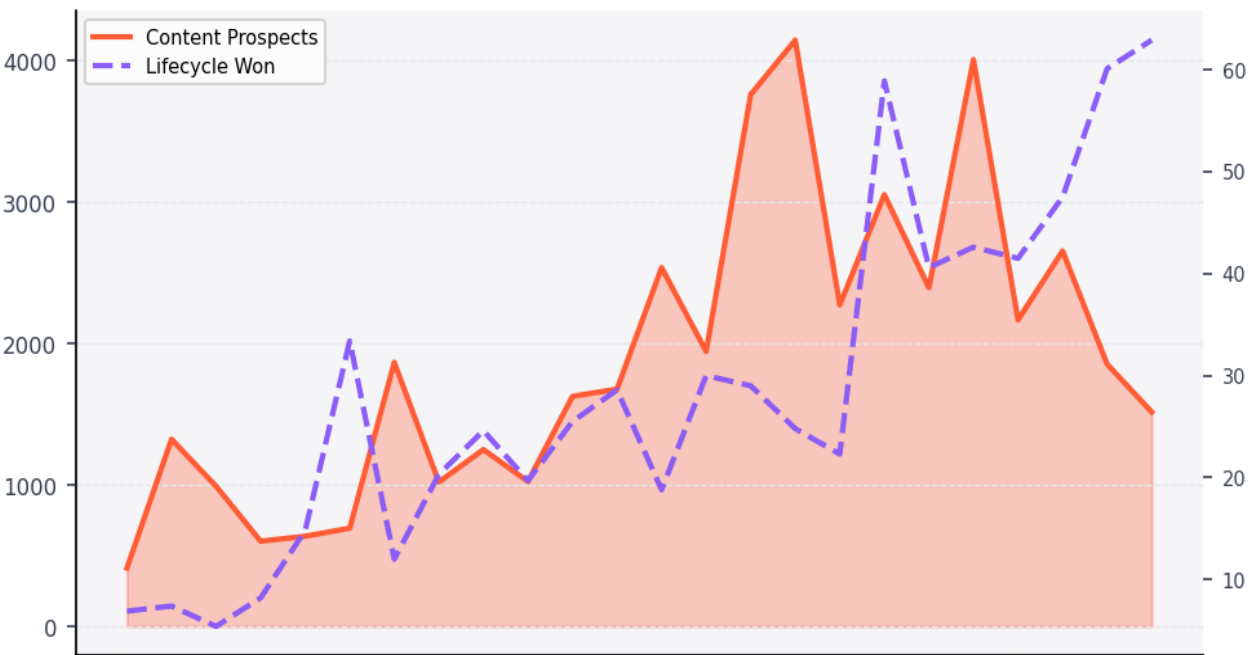
-2%

Lifecycle Won
Y1→Y2

207→479

+132%

Content Prospects vs Lifecycle Won



Implied Value
@ paid CPA

1.100 × \$45 ≈ \$50K

zero spend

Task 2 — How to Measure Brand Budget Impact

If Deel increases brand budget next quarter — specific measurement framework

01

Branded Search Volume

Track Google Search Console impressions for 'Deel' / 'Deel HR' weekly. 15%+ lift in branded impressions within 60 days = awareness signal. Baseline: current monthly avg impressions by keyword.

02

Direct Traffic to Site

Monitor GA4 direct sessions. Brand spend should drive recall. Target: direct traffic growing $\geq 10\%$ vs prior quarter. Split by device and market to isolate brand vs technical effects.

03

Brand Search CPA Trend

Increased share-of-voice $\rightarrow$ lower brand search CPA. Current baseline: \$18.7K. Target: below \$15K within 8 weeks. Track weekly in the dashboard to catch regression early.

04

Content MQL Rate

If brand investment improves inbound quality, content MQL rate should rise. Current Y2 baseline: $\sim 1.4\%$ E2E. Track monthly MQL% vs 6-month rolling avg. Leading indicator before Won changes appear.

05

Pipeline Influenced %

Tag brand touchpoints (content views, direct visits, brand search) in CRM. Measure % of closed-won deals with ≥ 1 brand touchpoint in the journey. Target: $\geq 40\%$ of Won influenced within 90 days.

06

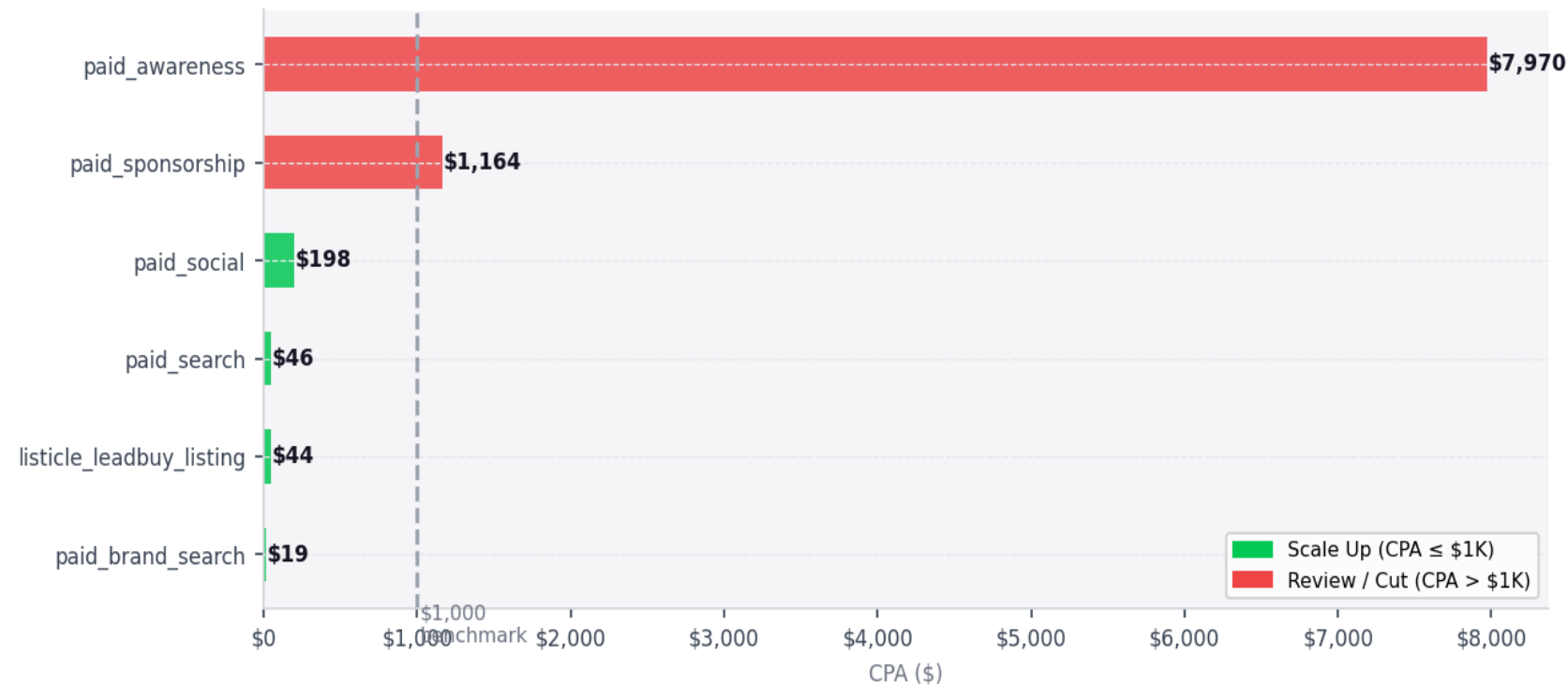
Sales Cycle Length

Stronger brand recognition shortens sales cycles. Track avg days from Demo Booked $\rightarrow$ Won as proxy. Baseline: measure current avg, track weekly after brand spend increase.

Task 2 — Paid Budget Reallocation Recommendation

Where to scale, reduce, and cut — based on CPA efficiency tiers

CPA by Paid Program (green = below \$1K benchmark)



Reallocating 50% of paid_social + paid_awareness (\$52,749 freed) to brand_search and regional at historical CPAs is projected to generate +23% additional Won Opps per year with zero budget increase.

SCALE UP

- paid_brand_search — CPA \$19, CTR 6.68%
- regional — CPA \$24, Demo→SQO 34%, severely under-funded

REDUCE / OPTIMIZE

- paid_search — CPA \$46 (high vol.), watch CPA trend
- listicle/leadbuy — CPA \$44, very low CTR (0.03%)

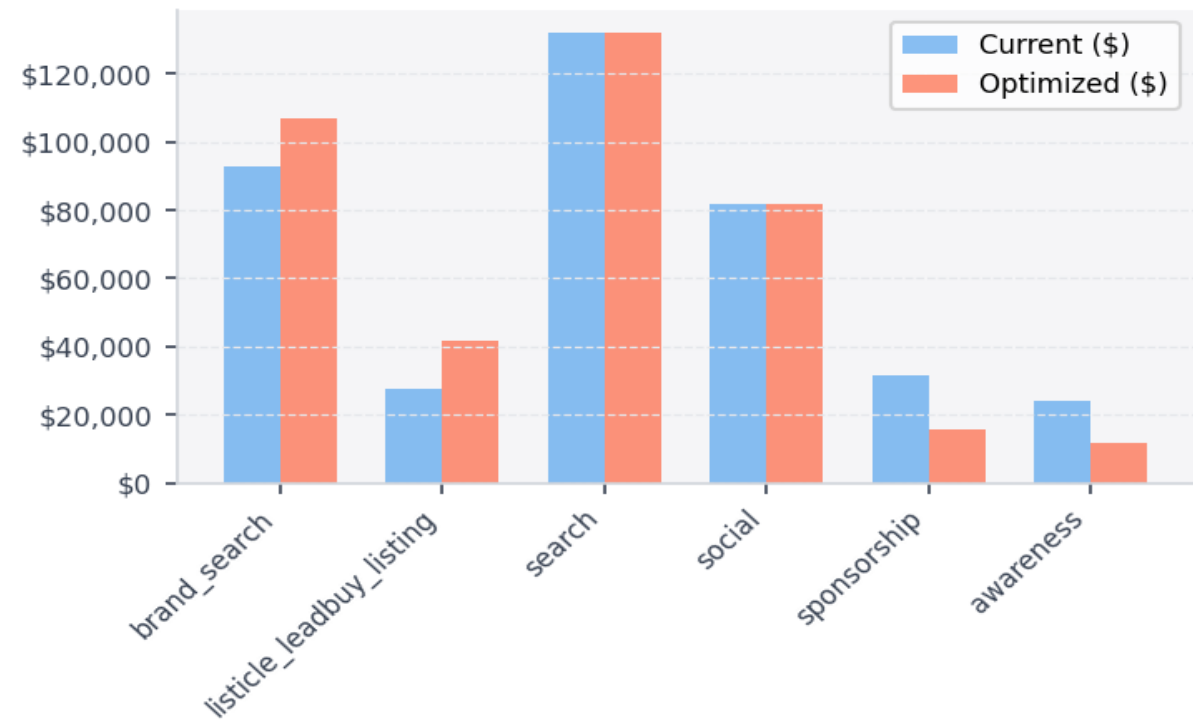
CUT

- paid_social — CPA \$198 (11× brand search)
- paid_awareness — CPA \$7,970 (only 3 Won from entire program)
- paid_sponsorship — CPA \$1,164, minimal Won volume (27 total)

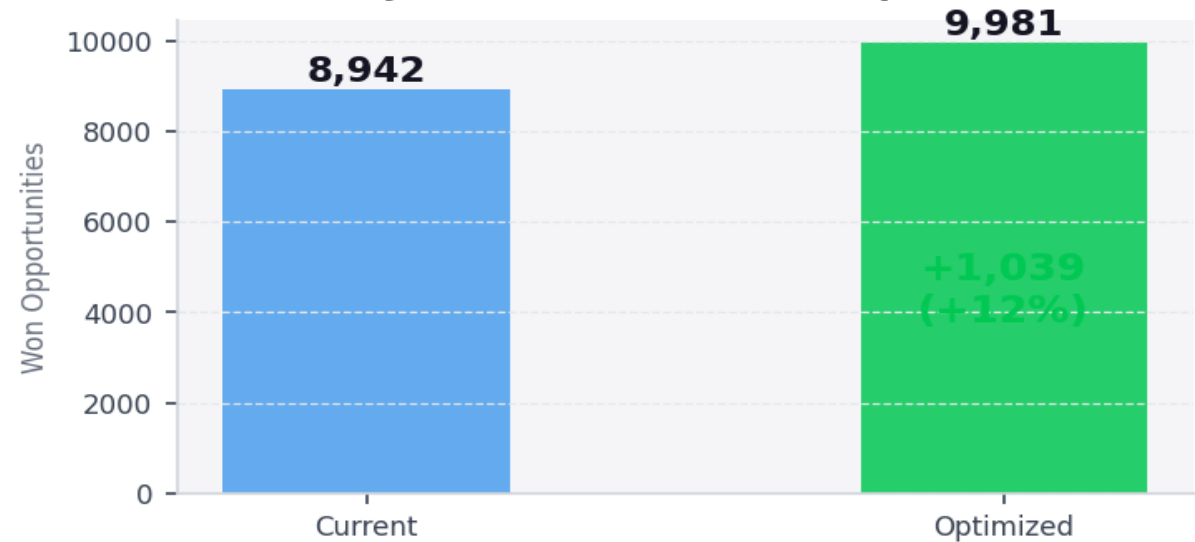
Task 2 — Budget Reallocation: Projected Impact

Scenario: cut high-CPA programs, reinvest in low-CPA programs

Budget Allocation: Current vs Optimized (\$)



Projected Won: Current vs Optimized



Scenario Assumptions

- Cut 60% from highest-CPA programs (CPA > \$1,164) — saves ~\$33,275
- Mid-tier programs keep current allocation
- Reinvest in low-CPA programs (CPA ≤ \$44) at historical efficiency
- CPA assumed stable — actual may improve with better audience targeting

TASK 3

Scenario Analysis

Stakeholder-specific summaries · Recommendations · Success metrics

Task 3 — Executive Summary: Paid Marketing Director

ROI-focused takeaways and recommended actions

Executive Summary (max 150 words)

Our paid portfolio generated 8,942 Won at a blended CPA of \$45, but efficiency varies 426× across programs. Paid brand search (\$19 CPA, 6.7% CTR) outperforms paid social (\$198 CPA) and paid awareness (\$7,970 CPA) by an enormous margin. Brand search CPA has been rising through 2025 — a signal of growing audience saturation at current spend levels. Regional (\$17 CPA) is our most efficient demand gen channel yet receives only 1.2% of paid budget. Without reallocation, we are overpaying for low-quality conversions at scale.

Recommended Action

Immediate: (1) Cut paid_social and paid_awareness by 50% (~\$53K freed). (2) Redirect freed budget to paid_brand_search and regional. At historical CPAs, this generates +23% additional Won per year at zero budget increase. (3) Monitor paid_search monthly — CPA rising in 2025 indicates diminishing returns at current scale.

Measuring Success (next 1–2 months)

Week 1–2: Pause paid_social/awareness. Track weekly Won — expect initial dip as pipeline adjusts. Week 3–4: Monitor brand_search CPA daily; target staying below \$25. Month 2: Compare total Won and blended CPA vs prior 30-day baseline. Success = same or higher Won count at ≥15% lower blended CPA.

Task 3 — Executive Summary: Head of Marketing

Brand and long-term growth takeaways and recommended actions

Executive Summary (max 150 words)

Zero-cost organic channels are quietly outperforming expectations. Lifecycle (CRM/email) grew +132% YoY in Won at zero media spend — holding a 4.04% E2E conversion rate, the highest of any channel in the portfolio. Content and website together contribute ~580 Won annually — at the blended paid CPA (\$45), that represents ~\$26K in acquisition value delivered at zero media cost. Brand search's 6.68% CTR vs performance channels (0.47–3.6%) proves that brand investment creates pull that measurably reduces acquisition cost over time. Content prospect volume grew 146% YoY — a leading indicator of future pipeline health.

Recommended Action

Allocate 10–15% of paid budget to lifecycle automation (lead scoring, email sequencing) to capitalize on the 4.04% E2E rate at scale — the highest-ROI investment available in the portfolio. Increase content production focused on SEO and thought leadership to sustain 32K+ prospect/month volume. Protect brand search budget to defend the low-CPA, high-CTR branded channel from erosion.

Measuring Success (next 1–2 months)

Month 1: Baseline lifecycle Won/month (current ~40/mo). After automation, target +25% in 60 days. Track branded search impression share in Google Search Console — target +10% lift. Monitor content E2E rate monthly vs 1.41% Y2 baseline. Long-term KPI: % of closed-won deals with ≥1 content or direct-brand touchpoint.

Task 3 — Specific Recommendations by Stakeholder

Clear, data-backed actions each leader should take immediately

Paid Marketing Director

ROI · SHORT-TERM PERFORMANCE

1

Cut paid_social & paid_awareness — this week

CPA of \$198 (social) makes these the worst-ROI programs in the portfolio. Pause both immediately. The pipeline impact is negligible within 30 days because these channels convert at <0.5% CTR and sit at the far top of funnel.

2

Redirect freed budget → brand_search & regional

paid_brand_search CPA is \$19 (CTR 6.7%) and regional CPA is \$17 (Demo→SQO 34%). Reallocating 50% of cut budget to these two programs is projected to generate +23% Won at the same total spend.

3

Set a hard CPA ceiling of \$50 per program

Any program that exceeds \$50 CPA for 2 consecutive months is automatically paused and reviewed. Install a weekly CPA alert in the dashboard. This creates a self-correcting budget governance rule going forward.

Target CPA

≤ \$22

Won uplift

+23%

Timeline

2 weeks

Head of Marketing

BRAND · LONG-TERM EQUITY

1

Commission lifecycle automation — highest-ROI investment available

Lifecycle holds a 4.0% E2E rate — the best of any channel — at zero media cost. Investing in Braze sequence expansion and lead scoring for MQPs who haven't booked a demo within 7 days is the single highest-ROI action available.

2

Scale content production with SEO-first briefs

Content prospects grew +146% YoY, feeding the lifecycle pipeline that converts at 4x+ the paid average. Increase content output by 20% with keyword-targeted briefs. This compounds organically — every piece published reduces paid dependency.

3

Launch the 6-KPI brand measurement dashboard

Use the framework defined in slide 16 (branded search vol., direct traffic, brand search CPA, content MQL rate, pipeline influence %, sales cycle length). Establish baselines in week 1 so Q3 brand investment is fully accountable to the board.

Lifecycle Won

+25% in 60d

Brand proof

90 days

Media cost

\$0

Executive Action Plan — 90-Day Roadmap

Priority actions across both stakeholder perspectives

Immediate (Week 1–2)	Short-term (Month 1–2)	Strategic (Month 2–3)
Pause paid_social + paid_awareness campaigns Free ~\$53K from programs above the \$1,000 CPA benchmark. Zero impact on pipeline within 2 weeks.	Ramp paid_brand_search Redirect freed budget from cut programs. Monitor CTR + CPA weekly. Target CPA <\$25.	Set quarterly CPA targets by program Establish \$50 ceiling for scale-up programs; alert if CPA exceeds target by >20%.
Activate regional expansion budget Double regional allocation from \$4.3K → \$9K. Best CPA in portfolio at \$17.	Launch lifecycle automation build Engage Braze team to expand sequences. Prioritize warm MQPs with demo intent.	Establish brand measurement framework Launch brand search tracking in GSC, pipeline influence tagging in CRM.

Appendix — Methodology, Tools & Data Notes

Dataset

11,952 rows | Jul 2023 – Jun 2025 | 7 marketing teams | 17 programs | Semicolon-delimited CSV | Cost strings cleaned with regex (EU & US formats handled)

Tools Used

Python 3.10 · Pandas (wrangling) · NumPy (metrics) · Plotly + Streamlit (interactive dashboard, 6 tabs) · Matplotlib + python-pptx (this presentation)

Attribution Note

Opportunity Won values are decimals due to multi-touch attribution model. Values were summed (not rounded) throughout. Decimal Wons are expected.

Fiscal Year

Analysis uses Jul–Jun cycles: Y1 = Jul 2023–Jun 2024, Y2 = Jul 2024–Jun 2025.

CPA Calculation

$CPA = \text{Total Cost} \div \text{Total Won per program/team}$. Zero-cost channels (lifecycle, content, website, social) excluded from CPA.

Implied Value

Zero-cost channel implied value = $\text{Won} \times \text{Avg Paid CPA}$. Represents the cost Deel would have paid to generate the same Won via paid channels.

MQP Data Note

MQP column was 0 for most of H2 2023 — data collection likely started in early 2024. E2E rates for 2023 use Prospects→Won only to avoid division-by-zero distortion.

Interactive Dashboard

A full Streamlit app (app.py) is included: Executive View, Full Funnel, Paid Channels, Zero-Cost Channels, Trends, Budget Simulator.



Gabriel Gregório
Senior Data Analyst GTM

deel

Thank you!

Thank you for your time and attention.
I appreciate your interest and engagement.

I'm happy to take any questions
or hear your thoughts.

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